Amir Ata Ghaffarian

CONTACT INFORMATION

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Homepage | GitHub | LinkedIn

ABOUT ME

Computer Science graduate, passionate about Artificial Intelligence, Machine Learning. Experienced in building AI/ML systems and conducting research for real-world applications.

EDUCATION

AmirKabir University of Technology (Tehran Polytechnic) **Tehran, Iran**
B.Sc in Computer Science *2020 – 2025*
GPA: 4.0/4.0 (18.97/20) — Last Year GPA: 4.0/4.0 (19.60/20)
Relevant Coursework: Machine Learning, Deep Learning, Artificial Intelligence, Data Structures & Algorithms, Information Retrieval

HONORS

- Ranked in top **0.3%** (464/150,000) in National University Entrance Exam (Mathematics).
- Ranked in top **5%** (3rd) among Computer Science students at Amirkabir University.

RESEARCH INTERESTS

World Models & RL, Robotics, Embodied AI, LLMs, Neuroscience, Computational Biology

EXPERIENCES

Research Intern
Okinawa Institute of Science and Technology – Japan *Feb 2026 – Present*
Currently a Research Intern at OIST's Genomics and Regulatory Systems Unit, working under the supervision of Prof. Nicholas Luscombe and Dr. Charles Plessy. My research lies at the intersection of AI and Bioinformatics, where I apply advanced Deep Learning architectures such as Self-Supervised Learning and Graph Neural Networks to decode genomic rearrangement patterns. By representing pairwise alignments through a multimodal approach, we aim to uncover structural variations at scale. I manage the end-to-end computational pipeline on an HPC cluster, utilizing Slurm and Singularity for seamless scalability and reproducibility.

DevOps engineer
ICT Group, Tehran, Iran *Aug 2025 – Jan 2026*
Supported enterprise infrastructure by deploying and troubleshooting containerized services on Kubernetes and building monitoring and centralized-logging workflows with Prometheus, Grafana, Filebeat, and the Elastic Stack.

Site Reliability Engineering Intern
Neshan *Jul 2025 – Aug 2025*
Completed an intensive bootcamp on Site Reliability Engineering, gaining hands-on experience in system administration, infrastructure automation, container orchestration, and monitoring systems.

Research Assistant
Center for Information Systems and Data Science *Jan 2025 – Jun 2025*
Conducted research on predictive maintenance for batteries and bearings using transformers and diffusion models under the supervision of Dr. Babak Khalaj and Dr. Mohammad Hossein Rohban.

Software Engineering Intern
Yektanet *Jun 2024 – Jul 2024*
Developed Dotanet Ad Server features (ad retrieval, auctions, click/impression tracking) with Go, Docker, PostgreSQL, and Kafka, ensuring scalability across multiple publisher websites.

Undergraduate Teaching Assistant
Amirkabir University *2022 – 2025*
Machine Learning (Prof. Ali Zarchi),
Advanced Programming, AI, Algorithms (Prof. Mehdi Ghatee, Amirkabir University),
Image Processing (Prof. Mostafa Shamsi, Amirkabir University),
Operating Systems, Software Engineering (Prof. Zahra Ghorbanali, Amirkabir University)

PROJECTS

Goal-Space Planning with Subgoal Models – Reproduction GitHub | Interactive Report
Built a tabular, from-scratch reproduction of Lo et al. (JMLR 2024) on FourRooms with exact subgoal models, covering 8 experiments (Python, NumPy). GSP halves the episodes Sarsa(λ) needs to reach near-optimal behavior (35 to 18). Documented three open discrepancies with the paper.

MyTorch – Deep Learning Framework GitHub
Built a from-scratch NumPy-based automatic-differentiation and neural-network framework with a PyTorch-like API, forward/reverse-mode AD, optimizers, training diagnostics, and reproducible run tracking. Used agentic coding partner and validated generated changes through 1,200+ Python tests, gradient checks, strict type/lint gates, multi-version CI, benchmarks, and detailed English technical documentation.

Knowledge Graph Generator GitHub : Developed a system to generate knowledge graphs from various data sources using Python. Integrated with FalkorDB for graph storage, Kafka for real-time data processing, MinIO for object storage, Redis for ensuring idempotency.

DiffRUL-CMAPSS GitHub
Built a diffusion-based data augmentation model (PyTorch) for aero-engine RUL estimation, integrating LSTM, transformers, and DDPM, based on a 2024 RESS paper.

Search Engine GitHub
Designed an IR system (Python, TF-IDF, ParsBERT) with K-Means clustering and FAISS for efficient document retrieval.

Dotanet GitHub
Created a scalable ad platform (Go, PostgreSQL, Kafka) with RESTful APIs for ad serving, auctions, and real-time tracking.

SKILLS

Programming Languages	Python, Golang, C/C++, Java
Frameworks & Libraries	NumPy, Pandas, TensorFlow, PyTorch, Scikit-learn, OpenCV , NLTK, FastAPI, Gin, Django
Databases	MySQL, PostgreSQL, Redis, FalkorDB, MinIO
Tools	Linux, Git, Docker, Docker Compose, Kubernetes, Slurm, Singularity, Kafka, Prometheus, Grafana, Nginx, Gitlab CI/CD, ELK Stack

CERTIFICATES & WORKSHOPS

Machine Learning Specialization DeepLearning.AI, Stanford University, Issued May 2024
Neuromatch Deep Learning Course Neuromatch, Issued Jul 2025
Foundational Cloud Practitioner Cloud Academy, Issued Sep 2025
Project-Oriented Course In Backend Development With Golang Quera, Issued Dec 2024
Task-Oriented Course in Linux Quera, Issued Mar 2024
Advanced Algorithmic Thinking and Data Structure Quera, Issued Oct 2023